DERIVED CATEGORIES OF COHERENT MODULES IDENTIFY WITH THE COHERENT SUBCATEGORIES ON A NOETHERIAN SCHEME

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Allowed standard prerequisites (editorial).

- The natural functor from the derived category of coherent modules to the derived category of quasi-coherent modules is fully faithful, and its essential image consists of the objects whose cohomology sheaves are coherent. invoked at proof lines 2830-2835 and 2839-2844. Supplies full faithfulness of the left vertical arrow and detects when an object of D^-(QCoh) comes from D^-(Coh).
- On a Noetherian scheme the derived category of quasi-coherent modules identifies with the derived category of all modules, preserving coherent cohomology sheaves. invoked at proof lines 2831-2833 and 2839-2842. Transports an object K of D^-_Coh to an object K' of D^-(QCoh) with the same cohomology sheaves.

Source reference. Original lemma 0FDA: sources/stacks-perfect/pinned-original-source.tex, source lines 2782–2782. Cited at proof lines 2830-2831 and 2842-2844 for full faithfulness of the left vertical arrow and for essential image detection; the square argument is inside the counted span.

Source reference. Original proposition 09T4: sources/stacks-perfect/pinned-original-source.tex, source lines 2026–2026. Cited at proof lines 2831-2833 and 2839-2842 for the bottom equivalence and the transport of K to K'; the essential surjectivity assembly is inside the counted span.

Proposition 0.2. *Let X be a Noetherian scheme. Then the functors*

$$D^-(Coh(\mathcal{O}_X)) \longrightarrow D^-_{Coh}(\mathcal{O}_X) \quad \text{and} \quad D^b(Coh(\mathcal{O}_X)) \longrightarrow D^b_{Coh}(\mathcal{O}_X)$$

are equivalences.

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Proof. Consider the commutative diagram

$$\begin{array}{ccc} D^-(Coh(\mathcal{O}_X)) & \longrightarrow & D^-_{Coh}(\mathcal{O}_X) \\ \downarrow & & \downarrow \\ D^-(QCoh(\mathcal{O}_X)) & \longrightarrow & D^-_{QCoh}(\mathcal{O}_X) \end{array}$$

By Lemma 0FDA the left vertical arrow is fully faithful. By Proposition 09T4 the bottom arrow is an equivalence. By construction the right vertical arrow is fully faithful. We conclude that the top horizontal arrow is fully faithful. If K is an object of $D^-_{Coh}(\mathcal{O}_X)$ then the object K' of $D^-(QCoh(\mathcal{O}_X))$ which corresponds to it by Proposition 09T4 will have coherent cohomology sheaves. Hence K' is in the essential image of the left vertical arrow by Lemma 0FDA and we find that the top horizontal arrow is essentially surjective. This finishes the proof for the bounded above case. The bounded case follows immediately from the bounded above case. □